



Systemic risk and oil price volatility shocks[☆]

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ARTICLE INFO

JEL classification:

G12

G3

Q4

Keywords:

Systemic risk

Oil price volatility

Supply-side shocks

Demand shocks

Idiosyncratic shocks

Supply chain network

Network centrality

ABSTRACT

We examine the impact of different types of oil price volatility shocks on firm's systemic risk using a large panel dataset of US firms. Oil price volatility shocks occur due to changes in supply or demand for oil, or through idiosyncratic fluctuations of oil prices. Our findings indicate that the supply-driven or idiosyncratic oil price volatility shocks reduce systemic risk, whereas demand-driven shocks have the opposite effect. Large-cap and high-beta firms amplify the impact of oil price volatility shocks on firms' systemic risk. Importantly, firms with extensive supply chain networks exacerbate systemic risk when facing demand-driven oil price volatility shocks.

The Great Financial Crisis, the turmoil in international capital markets in 2011, the COVID-19 recession are all associated with lower oil prices (Hofmann et al., 2023, p.2).

1. Introduction

In the aftermath of the Global Financial Crisis of 2007–08, the attention on systemic risk by financial regulators (a concept closely associated with the potential implosion of the financial system) has gained much prominence. It follows that, identifying the drivers of systemic risk is of immense importance to regulators and market participants (Acharya et al., 2012). To date, the majority of studies focuses

on the determinants of systemic risk in the banking sector (Adrian and Brunnermeier, 2016; Varotto and Zhao, 2018; Lin et al., 2023, see, among others), while only recently has there been an emerging part of the literature exploring non-financial sectors (Anginer et al., 2018; Dungey et al., 2022).

In parallel, numerous studies try to identify the determinants of various types of firm risk due to their impact on risk premium estimation, portfolio construction, hedging strategies, cost of capital measurement and corporate financial decisions, among others (see, for instance, Hamada, 1972; Robichek and Cohn, 1974; Andersen et al., 2005; Gaspar and Massa, 2006). Some common determinants are capital

[☆] Acknowledgements: We would like to thank the editor Prof. Iftekhar Hasan and the three anonymous reviewers for their useful comments and suggestions. We are also grateful to Andreas Kaeck, Richard Tol, Radu Tunaru, Carol Alexander, and Faek Menla Ali for their comments in an earlier version of the paper, which was presented at the Finance and Stochastics Seminar Series at the University of Sussex. We are also grateful to the participants of the Economics and Finance Seminar Series at the University of Portsmouth and the Economics Research Seminars at the University of Westminster for their valuable recommendations. In addition, special thanks to the participants of the 2021 National Conference of the Financial Engineering and Banking Society, the 2023 International Conference of the Financial Engineering and Banking Society, the 2023 International Symposium on Environmental and Energy Finance Issues Conference, the 2023 International Finance and Banking Society Conference and the 2023 International Conference on Sustainability, Environment, and Social Transition in Economics and Finance, for their helpful comments. All remaining errors are our own.

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<https://doi.org/10.1016/j.jfs.2025.101432>

Received 7 August 2024; Received in revised form 31 May 2025; Accepted 9 June 2025

Available online 28 June 2025

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structure, firm's fundamentals, corporate social responsibility, market competition, as well as, industrial production and GDP. Key among these risk factors is the contribution of each firm to systemic risk given its direct impact on economic stability.

In this study, we examine how oil price volatility influences the systemic risk of US firms over the period from 1991 to 2020. In the spirit of Kilian (2009), we decompose oil price volatility into its relevant shocks, considering the different sources that drive volatility, in order to investigate the effects of both demand- and supply-driven oil price volatility shocks. More importantly, though, the focal point of our study is to investigate the channels by which oil price volatility shocks trigger responses from the US firms' systemic risk. The reasoning behind this approach is to demonstrate that disaggregating oil price volatility shocks can yield more informative insights and provide a better understanding of the origins of systemic risk.

In order to elaborate on our arguments, we note that various measures exist for oil price volatility. Here, we consider three popular measures; namely, the realised, the conditional, as well as, the implied oil price volatility. Both realised and conditional volatility are current-looking measures of volatility, whereas the implied oil price volatility is a forward-looking measure. Because our study is primarily concerned with the role of oil price volatility shocks, we follow the empirical framework developed by Kilian (2009) in order to disaggregate oil price volatility (rather than oil price) into aggregate demand shocks, oil supply-side shocks and idiosyncratic shocks. In addition to this, we employ two measures of systemic risk, namely, $\Delta CoVaR$, and Marginal Expected Shortfall.

We place emphasis on oil price volatility due to its growing importance in recent years following the remarkable financialisation process of commodity markets and the subsequent strengthening of the link between oil and financial markets (see, indicatively, Tang and Xiong, 2012; Degiannakis and Filis, 2017). We attempt to provide new evidence in the existing literature on the complex interaction between oil markets and financial markets. This information is important for both policy makers and investors, as it provides insights with regard to the preparation of the financial system when systemic shocks occur in the market as well as portfolio construction and diversification.

It should be also mentioned that there some empirical studies focusing on the impact of oil price returns on systemic risk (see indicatively, Yin et al., 2021; Xie et al., 2023) or the effect of oil price volatility on variance risk premia (see for example, Bams et al., 2017) or systemic risk (see for instance, Khalif et al., 2021). Because our study is mainly associated with the origin of the shock, our study is also related to the strand of the literature that explores the link between oil price shocks and systemic risk (see indicatively, Qin, 2020; Maghyereh and Abdoh, 2021; Ouyang et al., 2022). Nevertheless, none of the latter studies consider the effects of the oil price volatility shocks on systemic risk. Hence the literature remains silent on the investigation of the relationship between two uncertainty indicators such as oil price volatility shocks and systemic risk.

To this end, we mention that the differentiation between oil price and oil price volatility matters, because the oil price shock represents an unanticipated change in oil price that affects the demand for firms' products as documented by Kilian and Park (2009), whereas an oil price volatility shock reflects an unanticipated fluctuation in the oil price variability that creates uncertainty and affects firms' planning and investment decisions as underscored by Chatziantoniou et al. (2021). Overall, oil price and oil price volatility shocks convey different information content in the attention of economic agents. In this respect, a thorough discussion about the existing literature can be found in the following section.

Hence, motivated by the recent financialisation of the oil market, as well as, by the strengthening of the nexus between oil and financial markets, we purport to evaluate the impact of different oil price volatility measures on the firms' systemic risk in the US market. The contribution of this study is twofold. First, from the standpoint of

empirical methods, by focusing on risk and looking into the impact from different oil price volatility shocks, we extend the framework initially developed by Kilian (2009) who focuses on different types of oil price (rather than volatility) shocks. We deviate from previous literature that entirely considers oil price shocks in various oil-related strands of the literature. Therefore, our study adds to existing literature in terms of introducing new evidence to test whether oil price volatility shocks could play a significant role in driving systemic risks. For instance, higher or lower systemic risks due to different oil price volatility shocks, could be beneficial for regulator's responses as they must apply more sophisticated policies. Indeed, a policy response to a systemic risk change due to a demand-driven oil price volatility shock should be different from the response to a supply-driven oil price volatility shock. This is due to the fact that different oil price volatility shocks could not necessarily trigger severe financial instability. In turn, it is not surprising if regulatory authorities do not follow a consistent policy over time. In the same line of reasoning, if different oil price volatility shocks are associated with higher or lower systemic risks, market participants' sentiment is affected differently, as not always widespread fears are rising. Overall, the source of the oil price volatility shock matters, given that the effects of demand and supply in the crude oil market on the US macroeconomic environment are qualitatively and quantitatively different.

Second, we employ a set of different approaches and statistics including baseline regressions, time-varying estimations, regressions analysis to explore time series and cross-sectional coefficients, market segment analysis, market portfolio analysis and supply chain network analysis. Therefore, our study provides possible channels through which systemic risk is amplified, that is for example, if firms are supply chain interconnected and highly risky. Although the majority of the systemic risk literature focuses on the banking sector, we do test whether systemic risk can also rise from different sectors in the economy in the light of commodity prices fluctuations. Overall, our study provides new evidence on the ongoing discussion between oil price volatility and financial markets.

To assess the aforementioned relationships, we employ an unbalanced panel data set of over 11,000 US firms. The main findings indicate a positive and significant risk exposure of the US firms' systemic risk measures to different types of oil price volatility measures (conditional, realised, and implied). However, the disaggregation of oil price volatility into three different shocks, i.e., supply-driven, demand-driven, and idiosyncratically driven, shows that demand-driven oil price volatility shocks lead to systemic risk increases, whereas supply-driven, and idiosyncratically driven oil price volatility shocks reduce systemic risk. In addition, our industry analysis lends support to our baseline findings. Moreover, our time-varying analysis implies that our results remain robust when we consider the relationship between different types of oil price volatility shocks and systemic risk at each point in time, including recent developments in the US oil market. What is more, portfolio analysis shows robust results and highlights the firm size, unsystematic and systematic risk as factors to generate systemic risks in the face of oil price volatility shocks. Finally, the supply chain network analysis indicates that interconnected firms which are mainly affected by demand-driven oil price volatility shocks tend to amplify systemic risk.

The remainder of the paper is organised as follows. Section 2 provides an overview of relevant literature. Section 3 describes the employed data and presents the construction of the oil price volatility, systemic risk measures and also the empirical methods. Then, the empirical findings are presented, interpreted, along with various robustness checks in Section 4. Next, the transmission channels are presented in Section 5. Finally, Section 6 concludes the study and provides recommendations and policy implications.

2. Review of the literature

Despite the fact that determinants of firms' volatility have been extensively studied in the literature, the effect of oil market developments on firms' systemic risk is a rather unexplored topic. However, as recently pointed out by Gao et al. (2022), heightened levels of oil price uncertainty could result in increased economic, market and business uncertainty and as such it could increase systemic risk. It is not a coincidence that during April 2020, when we experienced the worst historic downturn in the oil market with negative WTI prices, several articles in the financial press raised concerns about the potential bankruptcy of oil firms. Such failures could, in turn, create financial issues for the financial industry itself, which serves as the lender of these oil companies.¹

Furthermore, there are some additional studies that are associated with our motivation and therefore are worth mentioning at this point. For example, Reboredo (2015) investigates the impact of oil prices on the financial performance of renewable energy companies and reaches the conclusion that oil price developments play an important role when it comes to the systemic risk of this particular sector. Furthermore, Mensi et al. (2017) investigate systemic risk spillovers between four major stock markets and oil price returns and report that, there exists strong dynamic tail dependence, with oil exerting a stronger systemic effect on all four stock markets both in the short- and the long-run. Moreover, Yin et al. (2021) document that oil price returns have superior predictive power to forecast systemic risk spillovers in international markets and therefore oil price returns can be considered as an indicator of changes in systemic risk spillover.

More recently, empirical studies examine the effects of oil price shocks on systemic risk. For instance, Qin (2020) investigates the extent to which structural oil price shocks influence stress in financial markets at global level. The author finds that the impact of structural oil price shocks on financial systemic stress depends on the origin of shock, specifically whether the shock is supply-side or demand-side driven. In addition, the author documents that the impact is also different for countries that are net oil exporters or net oil importers. Furthermore, Maghyereh and Abdoh (2021) examine the GCC countries and conclude that oil supply shocks are the major factor to drive banks' systemic risk, while oil demand shocks appear more important during the global financial crisis and the COVID-19 pandemic.

Ouyang et al. (2022) examine the effect of oil price shocks on systemic risk of US stock returns and find that the negative impact created by negative oil price shocks on systemic risk is greater than the positive impact generated by positive oil price shocks. The authors also report that the impact of both positive and negative oil price shocks appears to be reduced during the COVID-19 pandemic.

With a particular focus on oil price volatility and systemic risk, Khalif et al. (2021) explore the impact of oil volatility on the systemic risk measures in major petroleum-based economies and document that increasing oil price volatility causes systemic risk increases. In addition, Xie et al. (2023) examine the connectedness of systemic risk between the oil and global stock markets. They find that oil market volatility causes systemic risk spillovers between oil markets and stock markets.

Other than the aforementioned studies, there is no empirical evidence on how the different oil price volatility shocks might impact systemic risk.² We posit that given the fact that oil price shocks exert different impacts on systemic risk (see, for instance Qin, 2020), we

should expect that oil price volatility shocks will trigger different responses in firms' systemic risk.

To motivate this link between oil price volatility shocks and systemic risk, we argue that although, oil price volatility is anticipated to have an amplifying impact on systemic risk (see, for instance Khalif et al., 2021), distinguishing between the different types of oil price volatility shocks would provide a more accurate picture of the interaction, uncovering hidden relationships among the variables of interest.

In particular, a positive supply-side development in the oil market such as technological advances in drilling which increases the oil supply is expected to reduce the price of oil. In response, oil price volatility associated with supply-side (SSV) rises, where firms benefit from such oil price decrease and thus they face less risks. So, we argue that a positive change in the world oil production is regarded as positive news for the corporate sector, and as such, the higher oil price volatility, leads to lower systemic risk.

Equivalently, a negative demand-side development such as decreasing demand for industrial commodities including oil, is expected to reduce its price, leading to higher oil price volatility that is associated with the demand-side, which, in turn, leads to higher systemic risk. We claim that a negative oil demand-side episode suggests that there is a weak and unstable global economic activity, which is associated with smaller business opportunities and hence, greater systemic risk.

Finally, a negative idiosyncratic episode such as reducing fears about future oil supply shortfalls and hence greater availability of oil for current use is expected to drive the oil price to fall. In response, oil price volatility associated with idiosyncratic-side increases, yet systemic risk is expected to fall. As the negative idiosyncratic shock lowers the uncertainty about the future availability of oil, which in turn reduces the uncertainty in the economy, creates a stable business environment and consequently leads to lower systemic risk.

Overall, our hypotheses on the responses of systemic risk to oil price volatility shocks are summarised in Table 1.

Another important feature of the systemic risk literature is the persistent attention to investigate potential determinants of systemic risk in the banking sector and particularly in the financial sector (Adrian and Brunnermeier, 2016; Varotto and Zhao, 2018; Lin et al., 2023, see, among others). In addition to this, there is a recent emerging attempt on the existing literature to explore non-financial sectors (Anginer et al., 2018; Dungey et al., 2022).

In this regard, we posit that it is rather important to investigate the non-financial firms in our context. Regardless of the fact that non-financial firms are not strictly part of the financial system, they are closely associated with the financial institutions via their financing and investment activities (see, Dungey et al., 2022). For instance, Hofmann et al. (2023) states that after the recent shift into a positive relationship between commodity prices and the US dollar (as opposed to their negative relationship in the previous decades), we experience a greater financial and macroeconomic instability as it is more difficult for economies to stabilise inflation and output, in tandem. Hence, such stagflationary effects, due to increased commodity price, may well lead to lower firm-level investments and higher costs. Overall, any factors that alter the systemic risk of non-financial firms due to their impact on their valuation, cash flows or revenues, could influence the systemic risk of the financial sector.

² When it comes to the difference between oil price and oil price volatility shocks, we argue that this differentiation matters because they convey different information content and thus cannot be regarded as identical variables. We use a correlation coefficient test to measure the degree of correlation between the oil (WTI) price returns and the oil price volatility measures. The results show that the correlation coefficients are negative, as expected due to the leverage effect, and equal to -0.43 (against OVX), -0.36 (against realised volatility) and -0.13 (against conditional volatility), which are rather low, indicating that they must not be considered as variables that convey the same information.

¹ See, for instance the article by CNN Business (2020) titled "Oil prices turned negative. Hundreds of US oil companies could go bankrupt". Link: <https://edition.cnn.com/2020/04/20/business/oil-price-crash-bankruptcy/index.html> and the article by Financial Times, (2020) titled "Bank investors confront a new fear: oil company defaults". Link: <https://www.ft.com/content/15687d3c-6248-11ea-a6cd-df28cc3c6a68>.

Table 1
Theoretical identification.

Impact	Channel		Expected coefficient
SSV rises (↑)	Increased oil supply	Drops the oil price	Lower systemic risk (↓)
ADV rises (↑)	Decreased oil demand	Drops the oil price	Higher systemic risk (↑)
IDV rises (↑)	Decreased idiosyncratic demand	Drops the oil price	Lower systemic risk (↓)
SSV drops (↓)	Decreased oil supply	Increases the oil price	Higher systemic risk (↑)
ADV drops (↓)	Increased oil demand	Increases the oil price	Lower systemic risk (↓)
IDV drops (↓)	Increased idiosyncratic demand	Increases the oil price	Higher systemic risk (↑)

Notes: This table shows the relationship between oil shocks and systemic risk. The Channel column shows the response of the oil price to a change in oil fundamentals of supply and demand. The Impact column denotes the response of the oil price volatility based on its origin (SSV, ADV, IDV) to a change in oil price. The Expected coefficient column indicates the reaction of firms' systemic risk to the change in oil price volatility. Overall, higher SSV, which results from lower oil prices due to higher oil supply, causes economic upturn; and thus, firms that are sensitive to SSV will see a decreases in their systemic risk. Higher ADV, on the other hand, which occurs when oil prices drop due to lower demand for oil, creates unstable global economic activity and it increases systemic risk. Finally, higher IDV, due to lower oil prices as the uncertainty of future availability of oil is lower, creates stable business environment, and therefore lower systemic risk. The reverse holds true in the bottom panel of Table 1.

To elaborate further, oil price volatility affects firms' strategic decision-making, cash flows and revenues, making them very cautious and hesitant with regard to making potential investments, in light of abrupt oil price fluctuations (see, [Elder and Serletis, 2010](#); [Henriques and Sadorsky, 2011](#); [Bloom, 2014](#)). Furthermore, [Gupta and Krishnamurti \(2018\)](#) identify oil price risk as a factor that has a strong impact on existing firm's cash flows and revenues. Thus, we argue that heightened oil price volatility might result in firms having to pay a higher risk premium as a compensation for increased levels of operational risk. Indeed, oil is a major input to production and therefore volatile oil prices are expected to affect the cost of producing goods and services and consequently firms' profitability and valuation. In turn, these adverse effects would lead to increased systemic risk by these non-financial corporations, which could transmit spillover effects to the financial sector.

Parallel to this, the vast majority of the oil-stock market literature has highlighted that oil market developments exert an impact across all sectors and firms of the economy (see, [Degiannakis et al., 2018](#), for a review of the related literature). For instance, studies such as those by [Jones and Kaul \(1996\)](#), [Huang et al. \(1996\)](#), [Wei \(2003\)](#) and more recently by [Giglio et al. \(2016\)](#), [Alaali \(2020\)](#) and [Gao et al. \(2022\)](#), to name a few, all show that oil prices and oil price volatility exert significant impacts on both financial and non-financial firms and as such, we anticipate that they should exert an impact on their systemic risk.

3. Dataset and methodology description

3.1. Sample

Our sample covers all NYSE and NASDAQ stocks in the Compustat database from 1990 to 2020. To be included, a firm is required to have available balance sheet items via Compustat. Initially, we download the daily stock returns of all available stocks. We find 11,448 firms with over 25 million daily observations. Then, we use monthly data using the last recorded price of each month, resulting in approximately one million firm-month observations. We choose a monthly frequency in our analysis for two main reasons. First, oil price volatility shocks can only be estimated in monthly frequency, and second firm risk measures need an adequate time horizon to be robustly calculated. For this reason, we drop the year 1990 from our analysis as the first year of each firm is used for the rolling window calculations. Overall, our sample spans across 11 industries for 30 years, while it is also unbalanced (see, [Table 2](#)). Hence, our final dataset contains 1,105,265 firm-month observations.

All firm financial data are downloaded from Compustat. The five factors for the US market, needed for the capital asset pricing models, are obtained from Kenneth R. French website. Oil related data are

Table 2
Year and Industry sample composition.

Panel A: Year composition				
Year	N. of firms added	N. of firms left	N. of obs	Sample %
1991	297	0	1009	0.09
1992	357	2	5912	0.53
1993	761	2	13,000	1.18
1994	627	38	20,858	1.89
1995	676	99	26,771	2.42
1996	908	114	35,588	3.22
1997	609	238	41,439	3.75
1998	558	331	45,438	4.11
1999	926	386	50,409	4.56
2000	572	511	52,455	4.75
2001	318	583	50,482	4.57
2002	225	475	47,516	4.30
2003	224	410	45,270	4.10
2004	313	299	44,848	4.06
2005	266	323	44,457	4.02
2006	293	337	43,800	3.96
2007	321	347	43,424	3.93
2008	137	373	41,726	3.78
2009	146	309	39,520	3.58
2010	303	294	39,209	3.55
2011	293	265	39,404	3.57
2012	384	296	40,465	3.66
2013	403	247	41,721	3.77
2014	401	284	43,224	3.91
2015	265	351	42,698	3.86
2016	185	348	40,938	3.70
2017	243	308	39,941	3.61
2018	223	277	39,461	3.57
2019	203	298	38,327	3.47
2020 ^a	11	2801	5955	0.54
Total	11,448		1,105,265	100.00
Panel B: Industry composition				
GICS	Industry name	N. of firms	N. of obs	sample %
	Others	190	4277	0.39
10	Energy	656	59,922	5.42
15	Materials	370	37,274	3.37
20	Industrials	1124	114,080	10.32
25	Consumer discretionary	1515	146,803	13.28
30	Consumer staples	375	35,937	3.25
35	Health care	1976	188,140	17.02
40	Financials	2270	235,092	21.27
45	Information technology	2270	210,285	19.03
50	Communication services	409	35,366	3.20
55	Utilities	67	5994	0.54
60	Real estate	226	32,095	2.90
Total		11,448	1,105,265	100.00

Notes:

^a In 2020, we did not find many control variables, this is why our sample drops considerably. Firms are allocated to industries based on Global Industry Classification Standard (GICS).

retrieved from EIA,³ and finally macroeconomic data are extracted from the FRED and OECD databases. Table A1 of the Online Appendix shows the variable definitions and the sources of data.

3.2. Oil price volatility estimates

In this subsection, we present the different oil price volatility measures that we employ in the study. Initially, we consider two current-looking measures, namely the conditional and realised oil price volatility, as well as, the forward-looking oil price volatility measure, which is the implied volatility index, OVX. Subsequently, we distinguish the source of oil price volatility, using (Kilian, 2009) methodology, into supply-driven, demand-driven, and idiosyncratically driven. All three volatility measures, as well as, the oil price volatility shocks are reported in monthly frequency for the period from January 1991 until December 2020.

For the estimation of the monthly conditional and realised volatility, we use the daily WTI crude oil prices for the period from 2 January 1991 to 31 December 2020. To estimate the oil price volatility shocks we further employ monthly data for the world oil production and Kilian's global economic activity index. The OVX is available from the CBOE and is also expressed in monthly frequency.

3.2.1. Realised oil price volatility

Motivated by Andersen et al. (2003), the monthly realised oil price volatility is calculated as the squared root of the sum of the daily squared oil price returns for any given month, as follows:

$$OILRV_t = 100 \sqrt{12 \sum_{s=1}^S (oilp_{t,s} - oilp_{t,s-1})^2}, \quad (1)$$

where $oilp$ denotes the WTI oil log-prices and S denotes the number of days (s) in any given month (t).⁴

3.2.2. Conditional oil price volatility

Regarding the monthly conditional oil price volatility, we employ a GARCH(1,1) model, which allows us to estimate the conditional variance, $\sigma_{t,s}^2$, of day s for any given month t . Having estimated the daily conditional variance of oil prices ($\sigma_{t,s}^2$), we compute the annualised monthly conditional volatility, as follows⁵:

$$OILCV_t = 100 \sqrt{12 \sum_{s=1}^S \sigma_{t,s}^2}. \quad (2)$$

³ The implied oil price volatility index (OVX) data are extracted from Bloomberg. This index is available from the Chicago Board Options Exchange (CBOE) since May 2007. We indicate that the inclusion of the OVX reduces our sample period and this is the reason of having 521,885 monthly observations when this index is employed. Potentially, the inclusion of this index does not capture events that took place prior to May 2007 and affected the oil market.

⁴ The choice of using daily prices to compute the monthly realised volatility is motivated by Wang et al. (2018) and Bucci (2020) and the fact that intraday data were not available to the authors.

⁵ The choice for the GARCH(1,1) was motivated by the fact that we needed a model that is able to capture the stylised facts of asset price volatility (e.g., volatility clustering) without adding further complexity to our modelling framework. In addition, our choice was also influenced by Hansen and Lunde (2005) who show that GARCH(1,1) is no inferior to other GARCH models for exchange rate volatility forecasting. However, we ran a set of different GARCH models, which we assessed based on the AIC criterion. The AIC tests suggested that the APARCH(1,1) was a more appropriate model. Subsequently, we used the extracted conditional volatility and we performed the remaining estimations. The results were qualitatively similar and as such, we report the findings based on the GARCH(1,1) model. The results are available upon request.

3.2.3. Oil price volatility shocks

Having estimated the monthly oil price volatility estimates, we employ the Structural VAR model by Kilian (2009) so as to identify the sources of volatility, as aforementioned. The Structural VAR model of order p takes the following general form:

$$A_0 y_t = c_0 + \sum_{i=1}^p A_i y_{t-i} + \varepsilon_t, \quad (3)$$

where y_t denotes the vector of endogenous variables, which are ordered as follows, world oil production, Kilian's global economic activity index and the oil price volatility. A_0 represents the $N \times N$ contemporaneous matrix and A_i denotes the $N \times N$ autoregressive coefficient matrices. The structural disturbances are denoted by the ε_t , and they are assumed to have zero covariance and to be serially uncorrelated. Hence, the variance-covariance matrix of these structural disturbances has the following form:

$$E[\varepsilon_t \varepsilon_t'] = D = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}. \quad (4)$$

The reduced form of the Structural VAR model (3) is obtained by multiplying both sides by A_0^{-1} , which yields:

$$y_t = a_0 + \sum_{i=1}^p B_i y_{t-i} + e_t, \quad (5)$$

where $a_0 = A_0^{-1} c_0$, $B_i = A_0^{-1} A_i$, and $e_t = A_0^{-1} \varepsilon_t$, i.e., $\varepsilon_t = A_0 e_t$. Note that the e_t are linear combinations of the ε_t , with a variance-covariance matrix of the form $E[e_t e_t'] = A_0^{-1} D A_0^{-1'}$.

In order to identify the structural disturbances, we are motivated by Kilian and Park (2009) and we impose the following short-run restrictions on A_0^{-1} :

$$\begin{bmatrix} e_{1,t}^{a\text{World Oil Production}} \\ e_{2,t}^{b\text{Global Economic Activity Index}} \\ e_{3,t}^{c\text{Oil Price Volatility}} \end{bmatrix} = \begin{bmatrix} \alpha_{11} & 0 & 0 \\ \alpha_{21} & \alpha_{22} & 0 \\ \alpha_{31} & \alpha_{32} & \alpha_{33} \end{bmatrix} \times \begin{bmatrix} \varepsilon_{1,t}^{SSV} \\ \varepsilon_{2,t}^{ADV} \\ \varepsilon_{3,t}^{IDV} \end{bmatrix}, \quad (6)$$

where SSV is the supply-driven volatility, ADV is the demand-driven volatility and IDV is the idiosyncratically driven volatility. More specifically, world oil production may not respond contemporaneously to shocks in Kilian's global economic activity index and changes in oil price volatility given the adjustment costs that are related to revisions in oil production. In addition, Kilian's global economic activity index may not react instantaneously to changes in oil price volatility. This is justified by the fact that within the same time point, demand for industrial commodities cannot be immediately adjusted to oil price volatility changes, given the slow adjustment in economic activity. Furthermore, world oil production is only contemporaneously affected by supply-side shocks, whereas Kilian's global economic activity index is instantaneously affected by supply-side shocks, and demand-side shocks. Finally, all types of shocks are allowed to contemporaneously influence changes in oil price volatility, given that asset markets (including the oil market) reflect immediately to information related to changes in demand or supply. Note that, all variables affect each other in their lagged relationship, suggesting that oil price volatility is anticipated to impact economic activity and oil supply, as shown by Guo et al. (2005), Rafiq et al. (2009) and Greenberg (2016).⁶

⁶ If volatility shows time dependence (e.g., autocorrelation) and exhibits large higher moments, a stationary bootstrap may be required. As a robustness test, we have used the stationary bootstrap method from Politis and Romano (1994), which accounts for serial dependence in the data by constructing

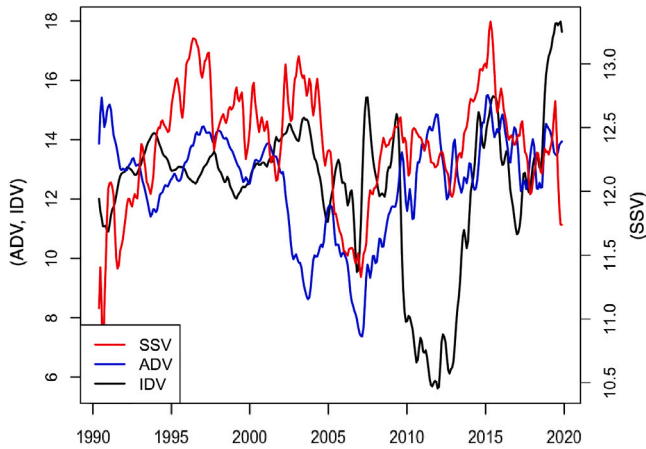


Fig. 1. Evolution of oil price volatility shocks over the years.
Notes: The Figure depicts the evolution of different oil price volatility shock time series.

Fig. 1 plots the visual representation of the three oil price volatility shocks, based on the Structural VAR model (5) with 8 lags, as suggested by the Akaike Information Criterion. We initially focus on supply-driven and the most prominent patterns. Thus, we observe a significant contribution of the supply-side to the oil price volatility during the first decade of our sample period, which is associated with Iraq's invasion of Kuwait in 1990 and the oil price spike as well as the significant oil price fall between 1997 and 1998 due to OPEC's increased quotas. Then, we notice a peak in 2016 associated with the oil price slump. Overall, we note that supply-driven oil price volatility shocks are typically associated with geopolitical unrest and OPEC's decisions related to production quotas.

Turning our attention to the most noticeable patterns of oil price volatility shocks related to demand-side, we observe the steady decrease in ADV from 2002 until 2007, which is a period associated by the unanticipated growth in demand for industrial commodities, including oil, by the emerging economies and especially China. Furthermore, we witness a peak of the ADV during 2014, which marked the end of a prolonged period when oil prices fluctuated between \$80 and \$100, during which the global economy was recovering from the global financial crisis. We argue that demand-driven oil price volatility shocks are endogenously generated during periods of economic boom or recessions.

Finally, the patterns for idiosyncratically driven oil price volatility shocks are mostly indicative of a persistent downward trend that reached a trough after the Great Recession. Furthermore, we observe that the main driver of oil price volatility during the period of the first wave of the COVID-19 pandemic in 2020 and the associated oil price crash is the IDV rather than demand-driven or supply-driven shocks.

Overall, the visual representation of the three oil price volatility shocks and the theoretical background that supports the patterns, provide evidence that abrupt declines in oil prices are accompanied by higher values of oil price volatility and vice versa.

3.3. Systemic risk measures

Our main measure of systemic risk is the $\Delta CoVaR$ measure proposed by Adrian and Brunnermeier (2016). The creators of this measure define it as “...the change in the value at risk of the financial system

conditional on an institution being under distress relative to its median state”. This measure is designed to capture the within-industry spillover effects in the financial sector due to a sudden change in the risk profile of a single financial institution. However, $\Delta CoVaR$ is a broad econometric technique that can be applied to any industry and any external shock that changes the risk profile of a given firm. In this paper, we focus on all firms comprising an economy and the source of the risk is the oil price volatility that affects the riskiness of a firm.

In our paper's context, systemic risk is defined as the contribution of Value at Risk (VaR) of one firm to the Value at Risk of the entire industry in which it operates. For example, higher oil prices can put American Airlines under distress, and this airline can transmit instability to the whole sector through price wars and other such spillover mechanisms. In this study, a firm under distress is reflected on the 95% of the losses distribution. This part of the distribution represents the highest monthly expected losses, which can be easily computed through the traditional VaR method. An alternative procedure to control for VaR , which is robust to outliers' spillover effects and is directly associated with systemic risk, is proposed by Adrian and Brunnermeier (2016):

$$\Pr(X_j | C(X_i) \leq CoVaR_{j|C(X_i)}^q) = q\%, \quad (7)$$

where X_j is the industry return losses ($X_j = R_j^{-1}$) conditional on the losses of a particular firm i ($X_i = R_i^{-1}$) at any part of the distribution (i.e., $q = 95\%$). Moreover, $CoVaR_{j|C(X_i)}$ is the Value at Risk of the industry j conditional on some event $C(X_i)$ of firm i . In this regard, $CoVaR$ can be implicitly estimated by running the following quantile regression:

$$X_{j,t}^q = a^q + \beta^q X_{i,t} + B^q M_{t-1} + u_{i,t}^q, \quad q \in (0, 1), \quad (8)$$

where the predictive values of $X_{j,t}^q$ are the Value at Risk of the industry conditional on $X_{i,t}$ at month (t). M_{t-1} is a vector that contains important macroeconomic variables lagged 1 period to capture the overall investment climate. Following Adrian and Brunnermeier (2016), we include market return losses ($X_m = R_m^{-1}$), the short-term market volatility and the daily yield from the US 10-year bond. Therefore, $CoVaR_{i,t}^q = \hat{X}_{i,t}^q$ and $CoVaR_{i,t}^q$ is the VaR of j conditional on VaR of i at any q and t given. Additionally, to demonstrate a more effective approximation of systemic risk, we use the $\Delta CoVaR$ measure, which is the change in $CoVaR$ of firm i at $q = 95\%$ to its median state ($q = 50\%$). The median state of any firm can be estimated by running Eq. (8) at $q = 50\%$ and then saving its fitted values ($CoVaR_{i,t}^{0.50}$). In other words, we run Eq. (8) twice at $q = 95\%$ and at $q = 50\%$, and save the fitted values. Then, $\Delta CoVaR$ can be measured as shown in Eq. (9):

$$\Delta CoVaR_{i,t} = CoVaR_{i,t}^{0.95} - CoVaR_{i,t}^{0.50}. \quad (9)$$

As a robustness, we use an alternative systemic risk measure, the Marginal Expected Shortfall (MES). Acharya et al. (2017) define MES as the short-run expected equity loss conditional on the market taking a loss greater than its Value-at-Risk at $1 - q$ (i.e., 5%). Practically, MES can be measured by estimating firm i 's losses when the market j as a whole is under distress. The MES reads:

$$MES_{i,t} \equiv -E_t(R_{i,t} | R_{j,t} \leq -VaR^{0.05}), \quad (10)$$

where $R_{i,t}$ and $R_{j,t}$ are the stock returns of the firm and market, respectively. Furthermore, $-VaR^{1-q}$ is a constant corresponding to our tail risk, which is chosen at 5% level in line with other studies (e.g., Cai et al., 2018; Raykov, 2022).

Fig. 2 illustrates the evolution of the $\Delta CoVaR$ and MES types of systemic risks. Specifically, it shows the impact of the Great Recession which is associated with a sharp increase in both types. It is worth noting that during this episode, the failure of large investment banks such as Lehman Brothers, posed substantial systemic risks. A notable observation is also related to the impact of the early stages of COVID-19 pandemic which reflects a sharp upward trend. In this respect, a

pseudo-samples from data blocks of randomly determined lengths. This approach has also been employed by Andrieu et al. (2022) to construct confidence intervals using stationary bootstrapping. Using this method, our main results remain unchanged and are available upon request.

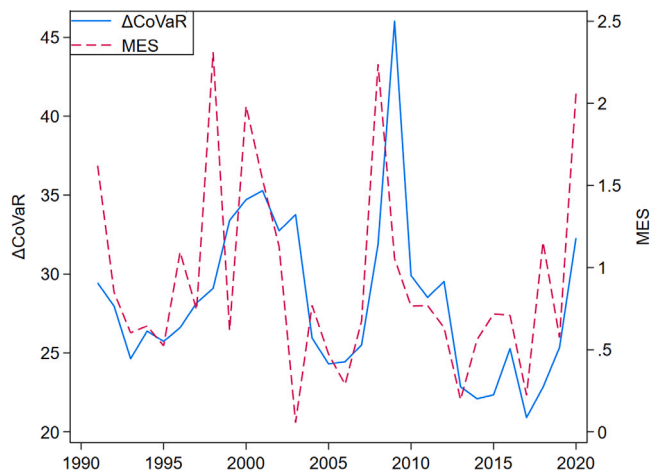


Fig. 2. Evolution of systemic risk over the years.

Notes: The Figure depicts the evolution of different time series. The time-series are calculated as the weighted average of all firms for any given year.

large systemic risk suggests a great speed of the COVID-19 pandemic spread according to which a failure of one firm might cause the failure of many.

3.4. Descriptive statistics

Table 3 reports the descriptive statistics of our variables. Starting with the oil-specific variables, the forward-looking implied volatility (OVX) appears to have a higher mean value and volatility than the current-looking volatility measures (i.e., OILRV and OILCV), as expected. Typically, a higher implied volatility value compared to the realised volatility indicates risk aversion in the market. Furthermore, implied volatility tends to be more volatile given that it incorporates expectations about rare events (which may not occur), whereas in the current-looking volatility measures such events are not incorporated in the estimation unless they have been materialised. On the other hand, the two current-looking volatilities share similar characteristics.

Regarding the oil price volatility shocks, it is evident that the idiosyncratically driven oil price volatility (IDV) is more volatile than supply-driven (SSV) and demand-driven (ADV) oil price volatility shocks. This can be attributed to the fact that idiosyncratic shocks are uncertainty shocks primarily generated by geopolitical turmoil, as well as, decisions made by money managers that participate in the oil market.

With reference to the stock market variables of $\Delta CoVaR$ and MES, they exhibit a leptokurtic distribution and positive skewness, as expected. In this regard, we notice that a long-right tail is indicative of a sluggish increase and a speedy decline. In addition, turning to the control variables which we explain in detail in the following paragraph, we report similar properties.

For firm-level control variables we use firm size (log of total assets, LNTA), leverage (LEV), earnings per share (EPS), dividend yield (DVY), book-to-market (B/M), liquidity ratio (LIQ), and the log of trading volume (LNVOLUME). Then, for macro-level data, we consider the TED spread (TED), volatility index (VIX), investment sentiment (CREDIT), consumer confidence index (CCI), business confidence index (BCI), and oil price returns (OILR) (see, for instance, White et al., 2015; Adrian and Brunnermeier, 2016; Güntner and Linsbauer, 2018). Note that OILR is included in the set of control variables so that we account for the leverage effect between the first and second moments of oil prices. Hence, the incorporation of OILR in the estimations allows us to isolate the effects of the oil price volatility shocks on systemic risk. A detailed

explanation of the variables can be found in the Online Appendix Table A1.

We include firm size as prior literature has shown that larger firms are better equipped to deal with systemic risk. We also control for variables that are found to explain high variation of the capital asset models, such as the previously mentioned firm-level variables. In addition to this, our macro variables are able to capture the fear in the market, the investment sentiment, the consumer sentiment, the business sentiment and consequently the overall economic climate. We also control for the time to capture different events in our long estimation period, and we also consider that different sectors have different risk levels.

4. Empirical findings

4.1. Baseline results on systemic risk

Our baseline regressions test the relationship between oil-specific variables (i.e., different types of oil price volatility and different types of oil price volatility shocks) on systemic risk, using the following specification:

$$y_{i,t} = a + \beta'X_t + \gamma'Z_{i,t} + \theta'M_t + \vartheta OILR_t + \Lambda + \varepsilon_{i,t}, \quad (11)$$

where $y_{i,t}$ is one of the firm-level systemic risk variables ($\Delta CoVaR$, MES), X_t shows the vector of oil price volatility shock variables (SSV, ADV and IDV), while we separately test for the oil realised (OILRV) and conditional (OILCV) volatility as well as the implied volatility (OVX). In addition, $Z_{i,t}$ is a vector that controls for various firm-specific characteristics and M_t is a vector of macro-level characteristics. We have also included the oil price returns ($OILR$) as an additional control variable so to capture the effects of the latter variable on systemic risk and leave any remaining effects to be captured by oil price volatility shocks. What is more, Λ shows different specifications controlling for firm (Firm FE), industry (Industry FE), year (Year FE), month-of-the-year (Seasonal FE), and month-by-year-by-industry fixed effects (S×Y×I FE). Finally, we cluster standard errors by firm.

We use standardised variables to obtain better distributional properties, to reduce the impact of outliers and thus to have more meaningful coefficients. We have standardised the risk measures and oil-specific measures to zero unity space. If control variables are not expressed in either ratio or a percentage, we use natural logarithms.

In this regard, Tables 4–5 summarise our baseline results. In particular, we note that each Table consists of seven columns. Left columns start with very general specifications (i.e., 1, 2 and 3) which are all related to the three types of oil price volatility and as we move to the right columns (i.e., 4, 5, 6 and 7) which are all associated with the three types of oil price volatility shocks our specifications become more restricted. Therefore, we follow a two step approach within each Table. More specifically, the first step is associated with the three types of oil price volatility such as realised, conditional and implied. We attempt to investigate their impact on systemic risk factors. For the second step, we seek to identify the specific sources of oil price volatility. In other words, to examine whether they are driven by supply and demand events in the oil market. On a final note, it should be mentioned that the second step analysis is only related to the conditional volatility measure.⁷

Our main results, with $\Delta CoVaR$ as a dependent variable, are reported in Table 4. The findings in columns 1, 2, and 3, show that the effects do not differentiate among the different types of oil price volatility, as positive and significant effects are reported. More importantly, we show that disentangling oil price volatility into its three shocks,

⁷ For brevity, we present only the baseline findings related to oil conditional volatility. We mention that the baseline results for the realised and implied volatility are qualitatively similar and further available upon request.

Table 3
Descriptive statistics.

Variable	Obs	Mean	Q1	Median	Q3	Std	Min	Max	Skewness	Kurtosis	Stationarity
Panel A: Oil variables											
OILR	1,105,265	0.0035	−0.0478	0.0146	0.0570	0.0863	−0.5681	0.5456	−0.7555	6.1780	−14.5227
OILRV	1,105,265	10.2839	7.2160	9.2129	12.0124	8.2929	3.2080	335.4480	27.3596	1056.6280	−11.0348
OILCV	1,105,265	11.4449	8.5277	10.3290	12.6943	8.5022	4.6050	272.3610	20.4801	576.2548	−7.8638
OVX	521,885	36.2901	28.1110	32.8213	41.3208	13.8559	15.6100	170.5500	2.0933	13.2778	−5.0852
SSV	1,104,265	12.6454	11.8469	12.6212	13.5062	1.3029	8.9537	16.2976	−0.1043	3.2286	−9.0202
ADV	1,104,265	12.5225	8.9865	12.4228	15.3285	6.0423	−5.0740	48.6557	1.2674	8.7874	−6.3817
IDV	1,104,265	12.2414	6.6475	11.3522	17.1643	8.2661	−6.0845	69.7510	1.0180	6.1184	−9.4128
Panel B: Stock market variables											
$\Delta CoVaR$	1,092,207	12.5179	9.3679	10.7543	15.3946	4.4121	5.6543	27.4241	1.1401	4.8924	1663.861
MES	1,074,777	1.0650	0.0000	0.0000	0.0000	5.6176	−5.2147	39.2829	5.3009	32.4081	1609.3549
Panel C: Control variables											
EIGEN	280,035	0.0121	0.0009	0.0051	0.0188	0.0153	0.0000	0.0941	1.6314	5.3517	−25.0572
LNTA	1,105,265	4.8556	3.2570	5.1907	6.7015	2.7246	−6.9078	13.9287	−0.7000	4.1512	16.907
LEV	989,062	2.2784	0.1355	0.4561	1.5495	5.6361	0.0045	42.0848	4.9728	31.6373	−25.4005
EPS	993,678	0.1469	−0.4500	0.0000	0.8800	1.8050	−7.1300	6.8000	−0.2194	7.7610	1.5068
DVY	1,104,560	0.2873	0.0000	0.0000	0.0800	3.0474	0.0000	440.0000	105.2143	14 355.5600	−50.4171
B/M	991,614	0.9703	0.3612	0.7222	1.2569	0.9920	0.0000	12.7921	3.2579	20.3488	2.8200
LIQ	843,697	4.6351	1.0499	1.8863	3.5075	110.4086	−12.7213	24 108.0000	177.2683	35 634.7500	29.3236
LN VOLUME	1,088,401	10.8506	9.3372	11.0589	12.5957	2.4617	−3.1355	20.0995	−0.4848	3.4211	454.257
TED	1,105,265	0.4758	0.2300	0.3700	0.5800	0.3668	0.1200	3.3500	3.0029	17.5314	7.7453
VIX	1,105,265	19.8202	13.8400	18.2300	23.9500	7.6492	9.5100	59.8900	1.6377	7.1728	5.6852
CREDIT	1,105,265	2.4379	1.8400	2.3400	2.8500	0.7507	1.3000	6.1000	1.5707	7.7202	2.0829
CCI	1,105,265	100.2792	99.3053	100.5938	101.3254	1.6151	96.2623	103.0603	−0.5393	2.7536	2.8203
BCI	1,105,265	99.8937	99.3127	99.9651	100.6066	1.1112	95.5984	102.2237	−0.8234	4.6671	18.7383

Notes: This tables reports the descriptive statistics. Panel A shows the oil related variables. Panel B describes the stock market related variables. Finally, Panel C shows the control variables. Column 1 reports the abbreviation of the variables as described in Table A1. Column 2 shows the available observations. Next columns show the mean, Q1, median and Q3 statistics. Then, the standard deviation, minimum and maximum of the variables are reported. We also report the skewness and kurtosis. Lastly, we report t-stat of the ADF test for stationarity (Panel A shows time series ADF and Panels B and C show panel ADF) and all variables appear to be stationary.

uncovers these hidden relationships that we referred to in Section 2. To be more explicit, SSV and IDV trigger negative and significant responses to the $\Delta CoVaR$, whereas the reverse holds true for the ADV. Arguably, this finding underlines that oil price volatility shocks can cause negative episodes in economic stability.

In turn, we assess whether our results remain similar when the MES is used as dependent variable. The findings are reported in Table 5. Even in the case of MES, we can observe that both SSV and IDV exercise a negative and significant effect on systemic risk, whereas the latter is positively impacted by the ADV. Thus, the baseline results shown in Tables 4 and 5 remain qualitatively similar and thus we confirm our argument regarding the link between oil price volatility and systemic risk.⁸

Our findings provide complementary evidence to those already documented in the existing literature. For instance, Ouyang et al. (2022) provide evidence that oil price shocks reveal different effects on the systemic risk of the S&P500 Index and the S&P500 Financial Sector Index when their origin is considered. Specifically, they find a positive (negative) effect of oil supply (oil precautionary demand) shocks on the $\Delta CoVaR$ and MES, while aggregate demand shocks do not exercise any impact on the two systemic risk measures. Similarly, Maghyreh and Abdoh (2021) show that oil price shocks have an impact on the $\Delta CoVaR$ and MES in the GCC region. In fact, oil supply shocks exert a higher impact due to oil production control, whereas aggregate demand shocks exercise a higher impact during periods of economic and financial uncertainty, such as the period of the Great Recession. Hence, our findings reveal new insights on the oil-systemic risk relationship, as we show for the first time that it is not only the oil price shocks, but also the oil price volatility shocks that exercise a significant and differentiating effect on systemic risk.

⁸ We conducted an additional test by using the *SRISK* methodology from Brownlees and Engle (2017) as an alternative systemic risk measure. *SRISK* extends the MES by incorporating a prudential capital ratio, which is primarily applicable to financial firms. Despite this limitation, we re-estimated our baseline regressions and found that the results remained consistent.

In retrospect, we argue that systemic risk behaviour is dependent on the type of the oil price volatility shock. Given that the uncertainty surrounding the oil market is considered as a major source of endogenous or exogenous shocks (depending on the industry), it has the potential to trigger simultaneous firm failures, leading to economic and financial instability.⁹ Nevertheless, our findings show that higher oil price volatility does not necessarily lead to greater systemic risk, as the source of oil price volatility matters. Such findings have not been previously reported by the related literature.

It would be instructive at this point to note that the baseline analysis refers to the total number of firms (energy and non-energy). In addition, our sample contains both oil users and non-oil users. Therefore, it deserves to re-estimate our baseline results excluding the energy firms, who are expected to show stronger response, as well as, splitting our sample between oil and non-oil users. Such analysis will provide additional insights as to whether different types of firms respond to oil price volatility shocks in a similar fashion or whether heterogeneous responses exist. We discuss this analysis in a greater detail in the industry analysis section, which serves as our first robustness test.

4.2. Time varying estimations

A first robustness test is related to the time-varying estimations using a rolling window approach. In a rolling regression the sample is dynamic over a given window and not fixed over the entire examined period. Empirically, the length of this window is often somewhere between 24 and 60 months, while there is some burning period (i.e., 30 months) in order to reach adequate number of observations, and so estimations to be statistically valid. The chosen window is shifted month by month over the entire sample period. From an econometric point of view, the use of rolling windows causes a sub-optimal use of the data

⁹ Kilian and Park (2009) document that oil prices are not exogenous in relation to the global economy. Economic shocks tend to drive macroeconomic fundamental including oil prices.

Table 4The impact of oil price volatility shocks on systemic risk ($\Delta CoVaR$).

Systemic risk ($\Delta CoVaR$)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
OILRV	0.441** (0.032)						
OILCV		1.839** (0.033)					
OVX			10.781** (0.065)				
SSV				−1.564** (0.021)	−0.800** (0.024)	−0.733** (0.023)	−0.791** (0.024)
ADV				0.681** (0.019)	1.424** (0.021)	1.379** (0.021)	1.425** (0.021)
IDV				−1.662** (0.023)	−0.145** (0.027)	−0.167** (0.027)	−0.134** (0.027)
LTNA					0.015** (0.003)	0.011** (0.004)	0.011** (0.004)
LEV					−0.005** (0.001)	−0.006** (0.001)	−0.006** (0.001)
EPS					0.006** (0.002)	0.002 (0.002)	0.002 (0.002)
DVY					−0.019 (0.017)	−0.033 (0.018)	−0.033 (0.018)
B/M					0.057** (0.005)	0.069** (0.005)	0.069** (0.005)
LIQ					0.024** (0.006)	0.026** (0.006)	0.026** (0.006)
LN VOLUME					0.000 (0.002)	0.004 (0.002)	0.003 (0.002)
TED					1.037** (0.016)	1.001** (0.016)	1.034** (0.016)
VIX					−2.400** (0.018)	−2.384** (0.018)	−2.410** (0.018)
CREDIT					2.833** (0.013)	2.880** (0.013)	2.824** (0.013)
CCI					−0.645** (0.051)	−0.546** (0.051)	−0.630** (0.051)
BCI					3.727** (0.041)	3.621** (0.041)	3.693** (0.041)
OILR					−1.162** (0.035)	−1.165** (0.035)	−1.167** (0.035)
CONS	29.189** (0.095)	29.079** (0.094)	24.783** (0.025)	30.365** (0.095)	27.618** (0.128)	27.482** (0.126)	27.639** (0.126)
Firm FE	Y	Y	Y	Y	N	Y	Y
Seasonal FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Industry FE	N	N	N	N	Y	N	N
$S \times Y \times I$ FE	N	N	N	N	N	N	Y
Obs	1,092,207	1,092,207	517,917	1,092,207	810,530	810,530	810,530
R ²	0.824	0.825	0.831	0.826	0.842	0.842	0.842

Notes: This table summarises the coefficient estimates of the linear panel data model, outlined in Eq. (11). All variables are described in Table A1. The sample period runs from 1991 to 2020. The cross section comprises a total of 11,448 companies, with a varying number of companies in each year. The model is estimated by means of the random and fixed effects estimation methods. Robust standard errors are indicated in round parentheses. Asterisks **, * denote the 1% and 5% levels of significance, respectively.

by picking an ad hoc window size rather than an optimal full sample. However, rolling window regressions can effectively capture short-term dynamic impacts of oil shocks on firm level, while controlling for various shocks that happened during these periods. We obtain monthly sample estimates of beta from rolling regressions. We use a 60 month estimation window to obtain timely estimates that pick up short-term fluctuations in betas.¹⁰ We estimate rolling sample betas by running the following time-series regression (Dangl and Halling, 2012; Cosemans et al., 2016):

$$y_{i,t,s} = a_i + \beta_{i,t} V_{i,t,s} + \epsilon_{i,t,s}, \quad (12)$$

where $y_{i,t,s}$ is the different firm risks and $V \equiv (X, Z, M, OILR, A)'$ is a vector that contains the explanatory variables, including the oil shocks similar to Eq. (11). The subscript $s=(1,2,\dots, \tau)$ is used to index the

y before the end of month τ which is the length of the estimation window, that is, $\tau=60$ months. The subscript τ is used to emphasise that we estimate integrated betas for each time using a rolling window of monthly data. The regression slope $\beta_{i,t}$ is our object of interest.

Fig. 3 depicts the time-varying effects for our main measure of systemic risk, that is, the $\Delta CoVaR$. Furthermore, it reports the percentage of firms that are impacted by each oil price volatility shock at each time point.

On the whole, we observe that SSV maintains its negative impact on systemic risk. More importantly, though, we observe that firms are susceptible to such oil price volatility shocks only in certain time periods, such as the early 2000s, 2016 and the first wave of COVID-19 pandemic. Hence, the SSV is more prevalent during periods of exogenous shocks, such as OPEC's efforts to coordinate production and geopolitical tensions.

Turning to the ADV, we show that its coefficient is mainly positive and that its impact on systemic risk of firms is mostly experienced

¹⁰ Alternative window lengths do not alter the main findings.

Table 5The impact of oil price volatility shocks on systemic risk: Alternative measure (*MES*).

Systemic risk (<i>MES</i>)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
OILRV	5.266** (0.199)						
OILCV		6.957** (0.207)					
OVX			22.020** (0.324)				
SSV				0.741 (0.723)	−0.316** (0.051)	−0.314** (0.052)	−0.316** (0.051)
ADV				2.457** (0.112)	1.282** (0.047)	1.282** (0.047)	1.274** (0.047)
IDV				−5.845** (0.137)	−1.895** (0.058)	−1.895** (0.058)	−1.885** (0.058)
LNTA					−0.266** (0.013)	−0.119** (0.021)	−0.118** (0.021)
LEV					0.030** (0.004)	0.032** (0.004)	0.032** (0.004)
EPS					−0.168** (0.011)	−0.137** (0.011)	−0.137** (0.011)
DVY					0.033 (0.085)	0.177 (0.096)	0.183 (0.096)
B/M					−0.377** (0.025)	−0.397** (0.030)	−0.396** (0.030)
LIQ					0.010 (0.029)	−0.040 (0.034)	−0.039 (0.034)
LN VOLUME					0.246** (0.011)	0.232** (0.012)	0.231** (0.012)
TED					2.161* (0.083)	2.177** (0.084)	2.169** (0.084)
VIX					2.379** (0.098)	2.380** (0.099)	2.369** (0.099)
CREDIT					−1.696** (0.072)	−1.700** (0.072)	−1.689** (0.072)
CCI					5.180** (0.272)	5.166** (0.274)	5.108** (0.274)
BCI					−10.110** (0.220)	−10.090** (0.221)	−10.058** (0.221)
OILR					−1.686** (0.077)	−1.697** (0.077)	−1.698** (0.077)
Cons	17.728** (0.556)	17.585** (0.556)	11.996** (0.121)	19.067** (0.562)	18.078** (0.675)	17.443** (0.676)	17.574** (0.676)
Firm FE	Y	Y	Y	Y	N	Y	Y
Seasonal FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Industry FE	N	N	N	N	Y	N	N
$S \times Y \times I$ FE	N	N	N	N	N	N	Y
Obs	1,074,777	1,074,777	517,917	1,074,777	810,530	810,530	810,530
R ²	0.017	0.017	0.036	0.019	0.028	0.026	0.025

Notes: This table summarises the coefficient estimates of the linear panel data model, outlined in Eq. (11). All variables are described in Table A1. The sample period runs from 1991 to 2020. The cross section comprises a total of 11,448 companies, with a varying number of companies in each year. The model is estimated by means of the random and fixed effects estimation methods. Robust standard errors are indicated in round parentheses. Asterisks **, * denote the 1% and 5% levels of significance, respectively.

during economic crises, when most firms within our sample are affected. More specifically, the positive impact begins after 2003 which is a period of cumulative positive oil demand-side developments in the price of oil and its volatility (see, Kilian, 2009). Finally, as far as the IDV is concerned, we report that its main negative effects on systemic risk appear during crises periods.¹¹ In this regard, we observe that during such periods of unprecedented high levels of business uncertainty, firms are more heavily impacted.

¹¹ We also proceed with two different tests to assess the impact of crises periods in the estimated coefficients. First, we split our sample to crises and non-crises periods. Second, we run interactions with crises dummies and oil price volatility shocks. Results reveal that SSV and ADV have identical signs to the baseline findings, which are intensified during crises periods, while IDV level term is positive during non-crises periods, whereas it turns negative during turbulent times, and overall confirm our previous estimations. The relevant results are presented in Table A4 in the Online Appendix.

Overall, the time-varying analysis allows us to extend our key findings, since, on the one hand, it confirms the baseline results from Tables 4 and 5, yet, on the other hand, it further shows that these impacts are observed at different time periods as the percentage of impacted firms provides evidence that not always oil price volatility shocks influence systemic risks.

Fig. 3 further includes the periods when important developments occurred in the US oil market. In particular, the green lines mark the beginning of (i) the US shale oil revolution, when technological advances in drilling from tight wells contributed to increasing oil production in the US since 2007; (ii) the new US oil transport infrastructure in 2012, when new pipeline, rail and barge projects enabled crude oil to flow to and from the Gulf of Mexico more easily; and (iii) the lift of the US federal export ban by the Congress in December 2015. These developments allowed greater US oil production and exports, which were beneficial for the US economy. It is rather interesting to note that once these developments took place, the impact of SSV and IDV on systemic risk diminishes. This is a rather important finding, given the

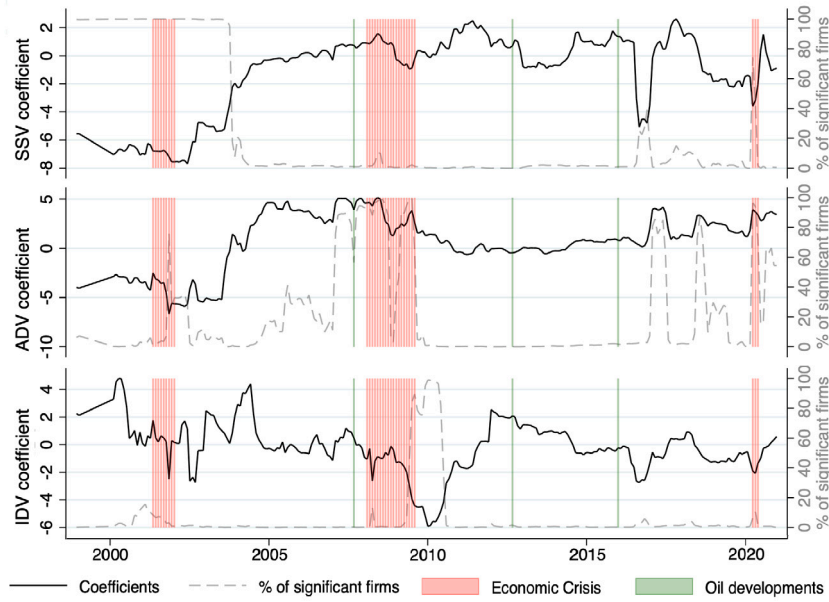


Fig. 3. Time-varying coefficients on systemic risk ($\Delta CoVaR$).

Notes: The figure shows the time varying coefficients of the three oil volatility shocks on systemic risk based on a rolling window of 60 month-observations. The first 60 observations are also a burning period. It also depicts the percentage of firms that are significantly affected by these oil shocks at 5% level. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

fact that these oil market developments in the US focus on the supply-side. Hence, such events should be anticipated to reduce the exposure of firms' systemic risk on oil price volatility shocks driven either by the supply-side or the uncertainty about the future availability of oil.

4.3. Time-series and cross-sectional coefficients

To understand the above findings, it is worth looking at the coefficients of the three oil price volatility shocks sorting by (i) firm and (ii) year. First, we run time-series OLS regressions per firm including all previous control variables and retain the coefficients. Second, we run cross-sectional OLS regressions per year and retain the coefficients.

Fig. 4 shows the distribution of the time-series coefficients per firm from the three oil price volatility shocks that are found to be significant at 5% level. In line with our theoretical identification, the vast majority of the SSV and ADV coefficients are negative and positive, respectively, indicating that the effects of SSV and ADV uniformly apply to all different firms. However, IDV coefficients appear to be both negative and positive. This may possibly be attributed to the fact that idiosyncratically driven oil price volatility shocks play an important role in periods of high levels of business uncertainty and such periods are not always evident in the US financial markets.

To further corroborate our results, we now visualise the cross-sectional coefficients per year in Fig. 5. Initially, SSV coefficients appear persistently negative with the highest magnitude at the beginning of the examined period. To this end, the existing literature indicates that it is the supply-side which plays a role in oil market developments during 1990s (see, Kilian and Murphy, 2014). Then, ADV coefficients change sign from negative to positive in early 2000, and the effect remains persistently positive and shows a higher magnitude until the end of our sample period, compared with the pre-2000 period. In this regard, Kilian and Murphy (2014) document that it is the demand-side that drives changes in the oil market after 2003. Lastly, IDV coefficients are positive and exhibit the highest magnitude in the beginning of the examined period and then turn to gradually negative but in a low magnitude after 2000 onwards. Once again, Kilian and Murphy (2014) show that it is the idiosyncratic component that appears to play a key role prior to 2000.

4.4. An alternative approach based on an intertemporal capital asset pricing model

Thus far, we regress systemic risk measures, which vary in time and firm, on oil price volatility shocks which only vary in time. Although, the results are rather informative on the effects of oil price volatility shocks on firm-level systemic risk, we should also assess the robustness of these findings, using an alternative approach that would capture the effects of firm-level sensitivity to oil price volatility shocks on firm-level systemic risk. Motivated by Bali et al. (2017), the suggested approach would be to construct firm-specific time-varying oil price volatility betas and subsequently test their impact on firm-level systemic risk. In other words, we seek to understand whether firms' exposure to oil price volatility shocks contribute to firm-level systemic risk in the cross-section and over time.

To achieve this, for each stock and for each month in our sample, we estimate the oil price volatility betas from the monthly rolling regressions of excess stock returns ($R_{i,t} - R_{f,t}$) on the oil price volatility shocks over a 60-month fixed window, while controlling for the five factors as proposed by Fama and French (2015) (also see Bali et al., 2017). Our expectation is that an Intertemporal CAPM model of this sort could provide cross-sectional and time-varying oil risk exposure factors:

$$R_{i,t} - R_{f,t} = \alpha_{i,t} + \beta_{i,t,SSV} SSV_t + \beta_{i,t,ADV} ADV_t + \beta_{i,t,IDV} IDV_t + \beta_{i,t,1} (R_{m,t} - R_{f,t}) + \beta_{i,t,2} SMB_t + \beta_{i,t,3} HML_t + \beta_{i,t,4} RMW_t + \beta_{i,t,5} CMA_t + u_{i,t}, \quad (13)$$

where the left part of the equation corresponds to the excess stock returns, while the right part consists of the SSV_t, ADV_t, and IDV_t variables, ($\alpha_{i,t}$) which shows the performance of a stock relative to the market portfolio, then the first factor ($R_{m,t} - R_{f,t}$) which is the excess return on the market portfolio, the second factor (SMB_t) which measures the return of small over large stocks, the third factor (HML_t) which represents the return of value stocks over growth stocks, the fourth factor (RMW_t) which reflects the difference of stock returns between robust and weak profitability firms, the fifth factor (CMA_t)

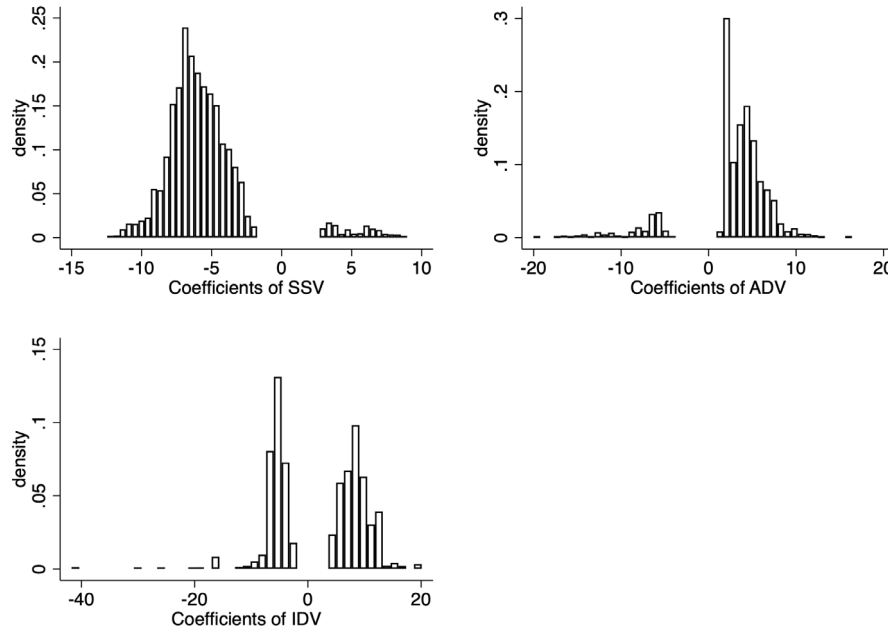


Fig. 4. Time-Series Coefficients (per firm) on Systemic Risk ($\Delta CoVaR$).

Notes: The figure shows the distribution of the coefficients of the three oil price volatility shocks on systemic risk ($\Delta CoVaR$). We run time series regressions per firm and we retain these coefficients. It depicts the significant coefficients of firms at 5% level.

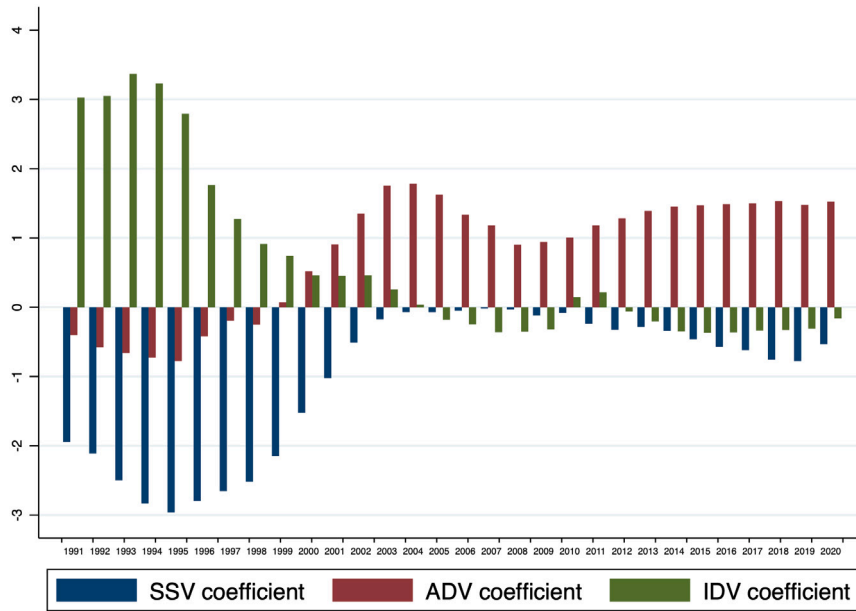


Fig. 5. Cross-Sectional Coefficients (per year) on Systemic Risk ($\Delta CoVaR$).

Notes: The figure shows the cross-sectional coefficients of the three oil price volatility shocks on systemic risk ($\Delta CoVaR$). We run cross-sectional regressions per year and we retain these coefficients. It depicts the significant coefficients at 5% level in any year given.

which signifies the return of low over high investment firms and $u_{i,t}$ denotes the residuals. Having estimated Eq. (13), we then re-estimate our baseline regression (Eq. (11)) as follows:

$$y_{i,t} = a + \beta' \mathbf{BV}_{i,t} + \gamma' \mathbf{Z}_{i,t} + \theta' \mathbf{M}_t + \Lambda + \varepsilon_{i,t}, \quad (14)$$

where $y_{i,t}$ is once again the firm-level systemic risk variable and $\mathbf{BV}_{i,t}$ now depicts the vector of the firm-level betas of oil price volatility shocks (i.e., $\beta_{i,t,SSV}$, $\beta_{i,t,ADV}$, $\beta_{i,t,IDV}$). We note that now the effect of oil price volatility shocks on the systemic risk varies both over time

and across firms. The remaining notation remains the same as in Eq. (11).

Panel A of Table A3 of the Online Appendix shows the estimated impact of the oil price volatility shocks on excess stock returns. We note that increased SSV and IDV lead to greater excess returns, which is rather expected, as we have already noted that increased supply-side and idiosyncratic-driven oil price volatility shocks denote positive news for firm activities. Similarly, the beta is negative for the ADV, which denotes the fact that increased demand-driven oil price volatility shocks

Table 6
The impact of beta oil shocks on systemic risk.

	<i>ΔCoVaR</i>				<i>MES</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
β_{SSV}	−0.057** (0.016)	−0.024* (0.010)	−0.057** (0.016)	−0.057* (0.016)	−0.002** (0.001)	−0.001 (0.001)	−0.001* (0.000)	−0.001* (0.000)
β_{ADV}	0.034** (0.011)	0.012 (0.007)	0.025* (0.011)	0.025* (0.011)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)
β_{IDV}	−0.004 (0.006)	−0.002 (0.004)	−0.002 (0.006)	−0.002 (0.006)	−0.001 (0.000)	−0.001 (0.001)	−0.001 (0.001)	−0.001 (0.001)
Controls	N	Y	Y	Y	N	Y	Y	Y
Firm FE	Y	N	Y	Y	Y	N	Y	Y
Seasonal FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Industry FE	N	Y	N	N	N	Y	N	N
$S \times Y \times I$ FE	N	N	N	Y	N	N	N	Y
Obs	795,250	590,470	590,470	590,470	786,181	583,637	583,637	583,637
R^2	0.847	0.865	0.865	0.865	0.014	0.026	0.025	0.024

Notes: This table summarises the coefficient estimates of the linear panel data model, outlined in Eq. (14). All variables are described in Table A1. The sample period runs from 1991 to 2020. The cross section comprises a total of 11,448 companies, with a varying number of companies in each year. The model is estimated by means of the random and fixed effects estimation methods. Robust standard errors are indicated in round parentheses. Asterisks **, * denote the 1% and 5% levels of significance, respectively.

are negative news and hence they lead to lower excess returns.¹² Panel B of Table A3 of the Online Appendix depicts the intertemporal hedging demand for low oil price volatility portfolios (for more information about ICAPM, see [Campbell, 2000](#)).

Based on the results from Eq. (13), we should expect the β_{SSV} and β_{IDV} to lead to lower firm-level systemic risk (since they represent positive news for the market), whereas the reverse should hold true for the β_{ADV} (since it represents negative news for the market), which would confirm the already established findings of the present study. The results from Eq. (14) are shown in Table 6.

Indeed, Table 6 shows that the effects of the firm-level oil price volatility shocks on firm-level systemic risk measure are as expected. In particular, we observe that the higher sensitivity in the supply-side and idiosyncratic oil price volatility shocks (high β_{SSV} and β_{IDV}) tends to lead to lower systemic risk, whereas, the reverse holds true for the firms' sensitivity to β_{ADV} . The results remain robust for both firm-level systemic risk measures, as well as, to the different specifications of Eq. (14). The only caveat is that the β_{IDV} does not retain its significant effect, as opposed to the significant IDV result shown in Tables 4 and 5. In any case, these results provide confidence that the initially estimated effects remain robust even when both the time and cross sectional effects are accounted for.

5. Transmission mechanisms

In order to observe tangible effects resulting from oil price volatility shocks on systemic risk, we next investigate the conditions that amplify these effects. We contend that, certain industries are more closely associated with oil price volatility, such as the energy sector and, as a result, are more likely to transmit heightened episodes of systemic risk. Similarly, firms with higher level of tangible assets (e.g., manufacturing firms) may also be more affected by oil price volatility shocks, which, in turn can lead to increased systemic risk. Moreover, firms with larger size and/or higher levels of systematic and unsystematic risk exposure may also have the potential to propagate larger systemic shocks throughout the economy. Lastly, firms that hold a central position in the supply chain and rely heavily on oil could also serve as significant transmitters of systemic risk. Next, we investigate the importance of such economic mechanisms.

¹² The reader can also observe the evolution of the weighted average oil price volatility betas in the Online Appendix, Figure A2.

5.1. Market segment analysis

In order to perform a market segment analysis, the first step is to exclude energy firms from the baseline analysis. As already mentioned, this choice is motivated by the fact that energy firms may influence the estimated coefficients that we have reported thus far. Table 7 reports the results which are related to the total sample period. Evidently, we note that the findings for all specifications tend to be consistent with those reported in Tables 4 and 5, in terms of the expected signs and significance. A plausible interpretation for this finding may possibly be attributed to the proportion of energy firms in our sample which is roughly estimated to 6% and therefore the weight of energy firms does not appear to exert a significant differentiation in our findings.

In Table 7 we also separate industries into (i) oil users, (ii) non-oil users, (iii) firms with higher level of tangible assets and (iv) firms with higher level of intangible assets, in order to thoroughly investigate our market segment analysis. We should mention that our sample is almost split equally between oil users and non-oil users, whereas the proportion of firms with higher level tangible (intangible) assets is 70 (30)%. Following [Kilian and Park \(2009\)](#), this motivation is attributed to the fact that heterogeneous responses may exist at the industry level to oil price volatility shocks, given the different degree of sensitivity varies across industries.

Interestingly enough, our results remain qualitatively similar in terms of the expected sign and significance, and this is observable in both measures of systemic risk. More specifically, the results show that SSV and IDV lead to lower systemic risk, whereas ADV aggravates it. Hence, the lack of any material difference in our sub-samples of firms results in the strengthening of our main findings, that the impact of oil price volatility on systemic risk depends on the source of the volatility. In this regard, we argue that our findings are not entirely unexpected.

On the one hand, oil is a primary raw material for both oil users (chemicals, manufacturing, or construction materials among others) and firms that mainly use tangible assets to cover the full spectrum of production (machinery, raw materials, and inventory). Therefore, both types of firms are vulnerable to oil price volatility shocks. For instance, oil price volatility shocks generated by oil price decreases due to increasing OPEC's quotas could have an impact on firm's inventory levels. In turn, this justifies why they are expected to be associated with uncertainties generated by oil price volatility shocks and this consequently raises systemic risks.

On the other hand, the fact that non-oil users as well as firms that mainly use intangible assets (assets without physical substance) which are both mostly financial services firms (such as banks and insurance firms), are affected by oil price volatility shocks, can be

Table 7
Market segments estimation results.

	EX-Energy firms		Oil users		Non-Oil users		Tangibles		Intangibles	
	$\Delta CoVaR$	MES	$\Delta CoVaR$	MES	$\Delta CoVaR$	MES	$\Delta CoVaR$	MES	$\Delta CoVaR$	MES
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
SSV	−0.750** (0.021)	−0.188** (0.054)	−0.78** (0.027)	−0.086 (0.068)	−0.713** (0.032)	−0.325** (0.087)	−0.716** (0.034)	−0.332** (0.075)	−0.745** (0.034)	−0.106 (0.071)
ADV	1.376** (0.017)	1.360** (0.049)	1.299** (0.023)	1.411** (0.063)	1.480** (0.027)	1.337** (0.077)	1.305** (0.031)	1.204** (0.069)	1.470** (0.031)	1.214** (0.064)
IDV	−0.167** (0.023)	−1.756** (0.043)	−0.144** (0.030)	−1.689** (0.058)	−0.190** (0.036)	−1.895** (0.064)	−0.075** (0.039)	−2.163** (0.086)	−0.287** (0.038)	−1.699** (0.079)
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Seasonal FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
$S \times Y \times I$ FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Obs	754,749	754,247	414,328	414,028	331,748	331,598	415,597	416,153	394,933	393,807
R ²	0.842	0.025	0.842	0.025	0.841	0.027	0.838	0.025	0.846	0.026

Notes: This table summarises the coefficient estimates of the linear panel data model, outlined in Eq. (11). All variables are described in Table A1. The sample period runs from 1991 to 2020. The cross section comprises a total of 11,448 companies, with a varying number of companies in each year. The model is estimated by means of the random and fixed effects estimation methods. Robust standard errors are indicated in round parentheses. Asterisks **, * denote the 1% and 5% levels of significance, respectively.

viewed through the prism of the financialisation of commodity markets (see, for example, Büyüksahin and Robe, 2014; Fattouh et al., 2013; Irwin and Sanders, 2011). In this regard, oil is used as a new category of financial assets within the portfolios of financial institutions such as brokerage firms, investment banks, and insurance companies since 2008. This practically implies that oil can be used by financial markets professionals for formulating effective portfolio diversification strategies. Consequently, this financialisation contributes to oil price abrupt fluctuations, which in turn could generate systemic risks.

In addition to this, oil extraction, oil production, and refining firms require a huge amount of capital and the financial sector is expected to act as their main lender to provide direct finance. In this regard, it is not surprisingly that financial firms are subject to oil price volatility shocks. So far, our market segment findings are somewhat in line with the evidence provided by Reboredo (2015), who reports that oil price dynamics (although the study does not focus on volatility) appear to exert around a 30% impact on the systemic risk of firms, with a particular focus on the renewable energy industry. In the following section we perform alternative tests in order to satisfy the robustness of our results.

5.2. Market portfolio analysis

Having established that oil price volatility shocks primarily matter for systemic risk analysis, we proceed with an additional channel based on a portfolio analysis. According to both theoretical and empirical predictions, large and highly volatile firms are sources of systemic risk (Adrian and Brunnermeier, 2016; Fernholz and Koch, 2017; Varotto and Zhao, 2018). Several studies (e.g., Acharya et al., 2017; Löffler and Raupach, 2018) discussed the connection between systematic and systemic risk. It is evident that a systematic risk can sufficiently jeopardise economic stability. In parallel, Acharya et al. (2012) and Benoit et al. (2017) underline that unsystematic risk is a result of correlated investments, which in turn causes contagion in the financial system. Therefore, firm-specific shocks can have significant macroeconomic consequences, and may harm economic stability. Based on this notion, we are splitting our sample across portfolios with different sizes and risk levels. Particularly, we consider size as well as unsystematic and systematic risk portfolios.¹³ Arguably, it would be interesting to investigate the channels through which oil price volatility shocks transmit systemic risk to the economy.

¹³ Refer to Section 1 of the Online Appendix for the construction of systematic and unsystematic measures.

In Table 8 we form quantile portfolios by virtue of (i) small to large firm size in Panel A, (ii) low to high unsystematic risk in Panel B, as well as (iii), low to high systematic risk in Panel C, and we examine the effects of oil price volatility shocks on systemic risk. For each month, we sort firms into quantile portfolios, where ‘low’ indicates the lower 20% of the firm size, unsystematic and systematic risk distribution. At the other end of the spectrum, ‘high’ indicates the top 20% of the distribution, while, ‘2’, ‘3’, and ‘4’ portfolios are formed accordingly.

Even though the majority of the results in Table 8 confirm our initial findings, it is rather interesting that the magnitude of the effects seems to be different in certain cases, depending on the quantile point. Starting with the firm size portfolios, we mainly observe that for larger firms, which tend to be more interconnected, the ADV (IDV) shows its highest positive (lowest negative) values. This finding suggests that indeed the firm size seems to affect the magnitude of the oil price volatility shocks to systemic risk. Turning to the unsystematic risk portfolios, the most interesting pattern is that there is clear evidence that SSV and IDV demonstrate their lowest negative values to firms which are characterised by high levels of unsystematic risk. This implies that firms facing high unsystematic risk are more prone to transmit higher systemic risk when facing negative SSV and IDV shocks. With reference to the systematic risk portfolios, we mainly notice that SSV seems to exhibit its lowest negative coefficient on portfolios with the highest level of systematic risk, which corroborates the findings from the unsystematic risk. Hence, the main picture that can be drawn from the latter portfolios is that firms with either high level of unsystematic or systematic risk can mainly amplify the SSV effects on systemic risk.

Overall, these results suggest that, regulators that aim to bring stability to the economy should not only be interested in the origin of the oil price volatility shock but also on the firm size, as well as, the levels of firm’s systematic and unsystematic risk.

5.3. Supply chain centrality

Our third transmission channel, which is a rather important novelty in this line of research, focuses on the relationship between a firm and its suppliers as a source of systemic risk. When firms receive shocks from oil price volatility changes, which create financial distress, we would expect these firms to transmit systemic risk episodes. It is important to note that the more central a firm is to its supply chain network, the more prevalent this transmission of systemic risk is expected to be. There is already a wealth of literature showing that the bankruptcy of a firm may well lead to financial distress or even bankruptcy of their customers or suppliers (Hertzel et al., 2008; Azizpour et al., 2018; Fang et al., 2023). We therefore analyse the relative importance of each customer or supplier in terms of its contribution to the networked

Table 8
Portfolio analysis: The impact of oil price volatility shocks on $\Delta CoVaR$.

Panel A: Ranking on firm size						
	1 LOW	2	3	4	5 HIGH	5 - 1
SSV	−0.676** (0.049)	−1.142** (0.045)	−0.812** (0.051)	−0.396** (0.060)	−0.439** (0.063)	0.237** [7.09]
ADV	1.400** (0.047)	1.119** (0.044)	1.383** (0.047)	1.360** (0.050)	1.650** (0.051)	0.250** [8.44]
IDV	−0.104 (0.059)	−0.145** (0.056)	−0.201** (0.059)	−0.182** (0.063)	−0.212** (0.059)	−0.108 [1.69]
Controls	Y	Y	Y	Y	Y	
Firm FE	Y	Y	Y	Y	Y	
Seasonal FE	Y	Y	Y	Y	Y	
Year FE	Y	Y	Y	Y	Y	
$S \times Y \times I$ FE	Y	Y	Y	Y	Y	
Obs	199,467	202,181	166,738	124,547	117,597	
R^2	0.825	0.835	0.843	0.857	0.861	
Panel B: Ranking on unsystematic risk						
	1 LOW	2	3	4	5 HIGH	5 - 1
SSV	−0.879** (0.060)	−0.059 (0.049)	−0.449** (0.047)	−1.325** (0.047)	−0.933** (0.053)	−0.054 [0.49]
ADV	2.215** (0.048)	1.198** (0.040)	0.931** (0.043)	0.989** (0.048)	1.495** (0.053)	−0.719** [63.35]
IDV	0.431** (0.052)	0.958** (0.048)	−0.026 (0.055)	−1.297** (0.063)	−0.800** (0.067)	−1.231** [150.4]
Controls	Y	Y	Y	Y	Y	
Firm FE	Y	Y	Y	Y	Y	
Seasonal FE	Y	Y	Y	Y	Y	
Year FE	Y	Y	Y	Y	Y	
$S \times Y \times I$ FE	Y	Y	Y	Y	Y	
Obs	87,800	153,608	181,488	192,386	193,514	
R^2	0.805	0.828	0.847	0.851	0.820	
Panel C: Ranking on systematic risk						
	1 LOW	2	3	4	5 HIGH	5 - 1
SSV	−0.608** (0.058)	−0.338** (0.057)	−0.664** (0.051)	−0.570** (0.050)	−1.307** (0.048)	−0.699** [68.5]
ADV	1.432** (0.054)	1.194** (0.050)	1.597** (0.046)	1.487** (0.044)	1.218** (0.047)	−0.214** [7.05]
IDV	0.006 (0.068)	−0.534** (0.064)	−0.467** (0.057)	−0.240** (0.053)	0.042 (0.059)	0.0036 [0.76]
Controls	Y	Y	Y	Y	Y	
Firm FE	Y	Y	Y	Y	Y	
Seasonal FE	Y	Y	Y	Y	Y	
Year FE	Y	Y	Y	Y	Y	
$S \times Y \times I$ FE	Y	Y	Y	Y	Y	
Obs	142,704	135,996	162,142	175,333	188,949	
R^2	0.827	0.853	0.851	0.853	0.836	

Notes: This table summarises the coefficient estimates of the linear panel data model, outlined in Eq. (11). In this table, we rank stocks into quantile portfolios of firm size, unsystematic, and systematic risk. All variables are described in Table A1. The sample period runs from 1991 to 2020. The cross section comprises a total of 11,448 companies, with a varying number of companies in each year. The model is estimated by means of the random and fixed effects estimation methods. Robust standard errors are indicated in round parentheses. Asterisks **, * denote the 1% and 5% levels of significance, respectively.

economy. Generally, from a supplier matrix A , one can compute a simple degree centrality vector, $\mathbf{c}$, as the simple sum of each row:

$$\mathbf{c} = A\mathbf{e}, \quad (15)$$

where $\mathbf{e}$ denotes a vector of ones. However, degree centrality is a simplified measure as it does not take into consideration the magnitude of each network. To overcome this problem, we use the eigenvector centrality (Bonacich, 1972). Eigenvector centrality is commonly employed in the finance research and it has several applications, such as measuring firms' CEO network, or supply chain network (El-Khatib et al., 2015; Wu, 2015; Conti and Graham, 2020). The notion of this measure is that each customer can be a supplier to other firms, and if the customer itself is an important supplier to many other firms, this implies that the importance of this customer in the economy is substantially increased. Instead of summing up all suppliers in a network, eigenvector centrality gives each supplier a weighted score based on the scores of its customers. Let c_i be the centrality measure for each firm i that is a supplier and let j be the customer at year t and let $w_{i,j}$ be the vertical relatedness between the two firms, (i.e., the

supplier i and the customer j). The eigenvector centrality of firm i is:

$$c_{i,t} = \left(\frac{1}{\lambda}\right) \sum_i \sum_j w_{i,t} A_{i,j,t} c_{j,t}, \quad (16)$$

where $\lambda \mathbf{c} = A\mathbf{c}$ as a matrix notation. Furthermore, λ denotes the largest eigenvalue of the supplier matrix A . The eigenvector centrality has an interesting property that it can be large either because a supplier (i) has many customers (j) or because it has important customers, or both. We normalise the measure to have comparable values across different years. To normalise it, we sum all firms in the network to N , $\mathbf{e}^T \mathbf{c} = N$, where N is the number of firms in the network. With this normalisation, the average centrality does not vary with the size of the economy for each year given.

We are interested in testing the triple relationship between systemic risk, centrality and oil price volatility shocks. To achieve this, we employ data by Frésard et al. (2020), who constructed a dataset with the pairwise vertical relatedness network (basic 10% granularity version; this is the 10% of all firm pairs with highest potential for vertical relatedness) for the period 1991–2020. There are more than 80 million pairwise connections in the data. To test econometrically the 3-way

Table 9

Ranking on systemic risk and Centrality: The impact of oil price volatility shocks.

Panel A: SSV coefficients and standard errors on ΔCoVaR						
Quintiles:		Ranking on systemic risk				
		1 LOW	2	3	4	5 HIGH
Ranking on Eigenvector centrality	1 LOW	−0.902** (0.219)	−1.244** (0.229)	−1.139** (0.277)	−0.906** (0.235)	−1.487** (0.234)
	2	−1.061** (0.211)	−0.598** (0.207)	−0.854** (0.236)	−0.242 (0.216)	−0.805** (0.223)
	3	−0.635** (0.198)	−1.009** (0.196)	−0.845** (0.225)	−0.780** (0.201)	−0.478** (0.207)
	4	−0.201 (0.204)	−1.127** (0.189)	−0.679** (0.211)	−0.526** (0.194)	−1.032** (0.216)
	5 HIGH	−0.567** (0.225)	−0.517** (0.182)	−0.920** (0.200)	−0.409** (0.187)	−0.474** (0.212)
Panel B: ADV coefficients and standard errors on ΔCoVaR						
Quintiles:		Ranking on systemic risk				
		1 LOW	2	3	4	5 HIGH
Ranking on Eigenvector centrality	1 LOW	1.080** (0.233)	1.555** (0.224)	1.144** (0.253)	1.562** (0.221)	0.331 (0.245)
	2	1.092** (0.202)	1.366** (0.185)	1.129** (0.210)	1.193** (0.194)	0.998** (0.221)
	3	1.454** (0.191)	1.396** (0.174)	1.540** (0.199)	1.467** (0.184)	0.999** (0.199)
	4	0.730** (0.196)	1.939** (0.170)	1.282** (0.189)	1.378** (0.168)	0.835** (0.199)
	5 HIGH	1.190** (0.191)	1.770** (0.157)	1.960** (0.174)	1.317** (0.157)	1.022** (0.181)
Panel C: IDV coefficients and standard errors on ΔCoVaR						
Quintiles:		Ranking on systemic risk				
		1 LOW	2	3	4	5 HIGH
Ranking on Eigenvector centrality	1 LOW	−0.194 (0.298)	0.240 (0.274)	−0.038 (0.313)	−0.382 (0.268)	−2.677** (0.318)
	2	0.428 (0.272)	1.067** (0.228)	0.697** (0.254)	−0.093 (0.232)	−1.219** (0.275)
	3	−0.032 (0.248)	0.788** (0.214)	−0.061 (0.236)	0.064 (0.220)	−1.123** (0.256)
	4	−0.027 (0.257)	0.752** (0.206)	0.187 (0.221)	0.216 (0.200)	−1.074** (0.253)
	5 HIGH	−0.234 (0.254)	0.672** (0.190)	−0.432** (0.206)	−0.432** (0.187)	−1.221** (0.232)
Controls	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Seasonal FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
$S \times Y \times I$ FE	Y	Y	Y	Y	Y	Y

Notes: This table summarises the coefficient estimates of the linear panel data model, outlined in Eq. (11). In this table, we rank stocks into quantile portfolios of systemic risk and eigenvector centrality. All variables are described in Table A1. The sample period runs from 1991 to 2020. The cross section comprises a total of 11,448 companies, with a varying number of companies in each year. The model is estimated by means of fixed effects estimation methods. Robust standard errors are indicated in round parentheses. Asterisks **, * denote the 1% and 5% levels of significance, respectively.

relationship, we first create a 5×5 portfolios ranked by systemic risk and centrality.

Looking at the panels of Table 9, it becomes obvious that with reference to the ADV, we document that the coefficients remain positive and significant for all different double sorted portfolios, and notably the highest ADV coefficients are, on average, in high centrality and high systemic risk quintiles. It is rather evident that the contribution of supply chain network to increasing systemic risk is generally weak in the case of SSV and IDV compared to ADV. The weak evidence regarding the importance of SSV and IDV may be attributed to the aforementioned major developments in the US oil market. Such oil supply-side developments might reduce the concerns of US firms to crude oil disruptions as they contributed to increasing flows of crude oil production.

Finally, in Table 10 we interact centrality with the three different oil price volatility shocks, while we also consider the full sample as well as sub-samples excluding energy firms, oil-users, non-oil users, firms that use tangible and intangible assets. Our results seem very robust as the interactions between EIGEN×SSV are insignificant, showing that

centrality is unimportant for supply-driven oil price volatility shocks. In contrast, the interaction EIGEN×ADV is always positive and highly significant in the full sample, the sample in which energy firms are excluded, the oil-users sample, and the sample of firms that employ tangible assets, which indicates that the more positive the ADV becomes, the higher the impact of supply chain network on systemic risk. In a similar vein, EIGEN×IDV appears negative and significant for the full sample, the sample in which energy firms are excluded, and the oil-users sample, which implies that supply chain network has a greater effect on systemic risk as the IDV becomes more negative. In addition, there is one notable observation. The results for the sub-samples of non-oil users and firms that employ intangible assets appear insignificant for all types of oil price volatility shocks. Reasonably, an intangible firm with a large supply chain network may not be able to cause systemic risk episodes.

6. Conclusion

This empirical paper uses an unbalanced panel data set of over 11,000 US firms for the period from 1991 to 2020 and makes use of

Table 10The impact of oil price volatility shocks and supply chain centrality on $\Delta CoVaR$.

	Systemic risk ($\Delta CoVaR$)		EX-Energy Firms	Oil-Users	Non-Oil Users	Tangibles	Intangibles
	Full sample						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
EIGEN	−1.009 (0.652)	−0.230 (1.837)	0.449 (1.914)	1.616 (2.081)	−4.427 (6.325)	−2.597 (2.603)	1.619 (2.661)
SSV	−0.949** (0.041)	−0.928** (0.050)	−0.970** (0.052)	−0.921** (0.061)	−1.134** (0.106)	−0.935** (0.068)	−0.911** (0.074)
ADV	1.328** (0.037)	1.257** (0.046)	1.222** (0.049)	1.182** (0.057)	1.313** (0.102)	1.127** (0.063)	1.393** (0.068)
IDV	−0.037 (0.046)	0.020 (0.056)	0.063 (0.060)	0.136 (0.070)	−0.128 (0.126)	0.197* (0.077)	−0.204* (0.082)
EIGEN × SSV		−1.613 (2.260)	−2.235 (2.336)	−3.059 (2.548)	−2.541 (7.723)	1.963 (3.152)	−5.444 (3.253)
EIGEN × ADV		4.858** (1.850)	4.871* (1.931)	6.338** (2.110)	−0.011 (6.274)	5.660** (2.877)	3.323 (2.476)
EIGEN × IDV		−3.668* (2.106)	−3.212* (1.793)	−5.753** (1.997)	9.381 (7.048)	−4.782 (3.255)	−1.540 (2.787)
Controls	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y
Seasonal FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
$S \times Y \times I$ FE	Y	Y	Y	Y	Y	Y	Y
OBS	248,322	248,322	218,437	171,699	43,263	120,419	127,903
R ²	0.858	0.858	0.857	0.858	0.849	0.855	0.862

Notes: This table summarises the coefficient estimates of the linear panel data model, outlined in Eq. (11). All variables are described in Table A1. The sample period runs from 1991 to 2020. The cross section comprises a total of 11,448 companies, with a varying number of companies in each year. The model is estimated by means of the random and fixed effects estimation methods. Robust standard errors are indicated in round parentheses. Asterisks **, * denote the 1% and 5% levels of significance, respectively.

monthly data to examine how oil price volatility shocks affect systemic risk. Within the framework of our study, we document that assessing the impact of oil price volatility on systemic risk at an aggregate level cannot uncover the full story. Thus, we emphasise on the importance of disentangling oil price volatility shocks by virtue of origin in order to delve deeper into this rather complex relationship.

The main findings document that, oil price volatility shocks exert a material impact on measures of systemic risk. Specifically, we provide evidence that supply-driven and idiosyncratically driven shocks reduce systemic risk, while a demand-driven oil price volatility shock increases systemic risk. Moreover, our results remain qualitatively similar when we perform an industry analysis for energy and non-energy firms, oil users and non-oil users, as well as firms that use tangible and intangible assets. Furthermore, our time-varying analysis confirms the important role of geopolitical and economic events as well as oil market developments to influence oil price volatility shocks and consequently systemic risk. In addition, our portfolio analysis confirms our baseline analysis and further stresses the importance of firm size, systematic and unsystematic risks as factors that affect systemic risk through the different types of oil price volatility shocks. More importantly, though, we highlight for the first time the role of supply networks to generate systemic risks when the interconnected firms are influenced by oil price volatility shocks, as shown in our supply chain network analysis. In particular, evidence suggests that the effects of demand-driven oil price volatility shocks on systemic risk are aggravated when firms are central to their supply chain network.

Our findings have important implications considering the fact that oil price volatility can be regarded as a leading macroeconomic indicator, and thus it conveys significant information in the attention of corporate decision makers, financial market participants and policy makers. More specifically, we refer to implications related to firm valuation, diversification opportunities and portfolio rebalancing, cost of capital, and policy initiatives for systemic risk.

As we show in the study, different oil price volatility shocks lead to changing levels of systemic risk. Hence, the information content advantage which characterises oil price volatility shocks creates initiatives in the attention of policy makers in order to manage and diminish systemic risk and thus, reduce the probability of an economy-wide instability event. More specifically, policy makers should monitor oil

price volatility shocks so as to gain time to prepare their responses for the protection of the economic system when such shocks occur in the market.

A promising area for future research may include additional international financial markets such as the Euro area, and Asian region industrialised economies, whose financial markets also play an important role in the global financial system. Moreover, another potential venue for further research would be a more in-depth examination and hence a deeper understanding of the link between oil price volatility shocks and systemic risk. Thus, it would be interesting to further investigate whether early warning systems that timely detect systemic anomalies in relevant industries could be predicated upon developments in the market for oil, which cause shocks to oil price volatility.

On a final note, we acknowledge the fact that our research relies on decomposing oil price volatility shocks into supply-driven, demand-driven, and idiosyncratically driven shocks, which is motivated by the original framework of Kilian (2009). We opted to use this framework so as not to add further complexity to our subsequent estimations, while maintaining our key thesis, namely, that the origin of the oil price volatility shocks triggers different effects to systemic risk. Nevertheless, future studies could employ either the more recent SVAR models to identify additional oil price volatility shocks, such as those suggested by Kilian and Lee (2014) and more recently by Cross et al. (2022). Even more, dynamic stochastic general equilibrium frameworks can be used to account for stochastic volatility shocks and level shocks in the macro economy that are important to model structural innovations of oil price volatility. Such frameworks would provide additional evidence in this rather important, but neglected, line of research.

CRedit authorship contribution statement

Ioannis Chatziantoniou: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Gonul Colak:** Writing – review & editing, Validation, Supervision, Conceptualization. **Michail Filippidis:** Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **George Filis:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Panagiotis Tzouvanas:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfs.2025.101432>.

Data availability

The authors do not have permission to share data.

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